

grow increasingly needy about customisation ability. And eventually as R&D progresses, the per unit manufacturing costs of these small production runs will be able to rival those of the traditional long running industries with large economies of scale.

Current research in 3D printing and their possible implications on several industries

3D printing has been evolving over the past 3 decades, and will continue to do so like all other technologies. Researchers are continuously breaking ground in terms of print size, speed and material integration. All of these developments create new solutions to existing problems, thus paving way for new entrants in the market and being responsible for an entire new breed of “world’s firsts”. There are systems being developed that can work by combining the benefits of both traditional machining devices such as CNCs and additive manufacturers such as 3D printers, thus removing the need for post-processing. This only leaves unlimited room for imagination as to the possible applications in different industries:



3D printed mechanical insect

- **Aircrafts and Aerospace:** Leading aircraft manufacturers such as Boeing and Airbus are using 3D printing to improve the performance by reducing the maintenance requirements and saving fuel costs with lighter parts. 3D printed aircraft components are 65% lighter than traditional aircraft parts but just as strong. How much does this save up to? To put things in perspective, for every 1Kg reduction in weight, airlines save around \$35000 in fuel costs spanning the lifetime of the aircraft. Aircraft designers already have in their sights manufacturing a whole plane by the years 2050. Boeing is actually somewhat of a pioneer in 3D printings having printed more than 25000 components being used in a variety of aircrafts. Components such as ECDs which require about 20 different parts to assemble due to its complex structure are now being manufactured as one piece. NASA is also taking advantage of cutting down on a long supply chain using 3D printers by trying to print products